

PERIODIC SIGNALS

ENERGY	POWER
$E = \infty$	$P = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s^2(t) dt \leftarrow \text{TIME DOMAIN}$
	$P = \sum_{n=-\infty}^{\infty} c_n ^2 \leftarrow \text{FROM COEFFICIENTS (PARSEVAL THEOREM)}$
	$P = \int_{-\infty}^{\infty} G(f) df$ $G(f) = \frac{dP}{df} = \sum_{n=-\infty}^{\infty} c_n ^2 \delta(f - f_0)$ POWER SPECTRAL DENSITY

APERIODIC SIGNALS

ENERGY	POWER
$\text{TIME DOMAIN} \rightarrow E = \int_{-\infty}^{\infty} s^2(t) dt$	$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} s^2(t) dt < \infty$
$\text{FREQUENCY DOMAIN} \rightarrow E = \int_{-\infty}^{\infty} S(f) ^2 df$	
$E = \int_{-\infty}^{\infty} G(f) df$ $G(f) = S(f) ^2$ ENERGY SPECTRAL DENSITY	